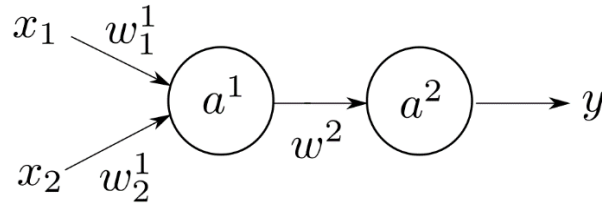


MLP Practice – Tutorial questions with solutions:

Q1. Let us consider a two layer neural network with the diagram



The neural network has two inputs x_1 and x_2 and one output y . The equations that represent this neural network are

$$a^1 = \phi_1(w_1^1 x_1 + w_2^1 x_2)$$

$$a^2 = \phi_2(w^2 a^1)$$

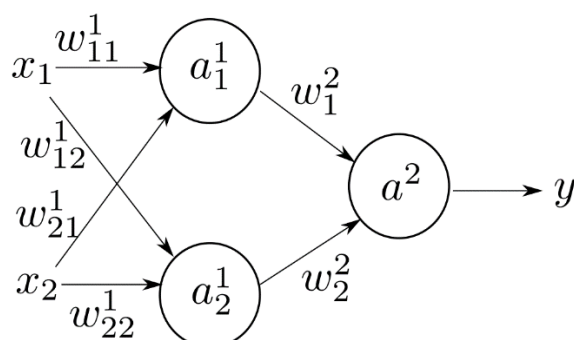
$$y = a^2$$

Where the first activation function is linear: $\phi_1(x) = \alpha x$, and the second activation function is a sigmoid $\phi_2(x) = \frac{1}{1+e^{-x}}$.

Let us consider the weights are $w_1^1 = 1$, $w_2^1 = 0.1$, and $w^2 = 0.1$, and the parameter $\alpha = 0.5$.

- 1) For each neuron calculate its induced local field and output for the following input value $x_1 = 1$, $x_2 = 2$.
- 2) If the desired output to the previous input is $d = 1$, use backpropagation algorithm to compute the gradient of the error with respect to the weights.
- 3) If the learning rate is 0.1, calculate the updated weights.
- 4) Calculate the output to the input value $x_1 = 1$, $x_2 = 2$ with the updated weights. Discuss the result.

Q2. We consider a neural network with two layers and the following diagram



The equations that represent this neural network are

$$\begin{aligned}
 a_1^1 &= \phi_1(w_{11}^1 x_1 + w_{21}^1 x_2) \\
 a_2^1 &= \phi_1(w_{12}^1 x_1 + w_{22}^1 x_2) \\
 a^2 &= \phi_2(w_1^2 a_1^1 + w_2^2 a_2^1) \\
 y &= a^2
 \end{aligned}$$

The activation functions $\phi_1(\cdot)$ and $\phi_2(\cdot)$ are sigmoid and rectified linear units, with the equations:

$$\begin{aligned}
 \phi_1(x) &= \frac{1}{1+e^{-x}} \\
 \phi_2(x) &= \begin{cases} x, & x \geq 0 \\ 0, & \text{otherwise} \end{cases}
 \end{aligned}$$

Note that this neural network does not include biases. The initial weights are $w_{11}^1 = 1$, $w_{12}^1 = 0.1$, $w_{21}^1 = 0.3$, $w_{22}^1 = 0.4$, $w_1^2 = 0.1$, $w_2^2 = 0.2$.

- For each neuron, calculate its induced local field and output for the following input value $x_1 = 1$, $x_2 = 2$.
- If the desired output to the previous input is $d = 0.5$, use the backpropagation algorithm to compute the gradients of the error with respect to all weights.
- Calculate the updated weights for a learning rate 0.1.

Q1 solution:

1) Induced local field and output neuron 1

$$v_1 = w_1^1 x_1 + w_2^1 x_2 = 1.2$$
$$a^1 = 0.60$$

Induced local field and output neuron 2

$$v_2 = w^2 a^1 = 0.06$$
$$a^2 = 0.5150$$

2) Backpropagation.

We first calculate the local gradients δ_1 and δ_2

$$\delta_2 = e_2 \phi'_2(v_2)$$
$$e_2 = d - y = 1 - 0.5150$$
$$\phi'_2(v_2) = \frac{e^{-v_2}}{(1 + e^{-v_2})^2}$$

Then,

$$\delta_2 = 0.1211$$

To calculate

$$\delta_1 = \phi'_1(v_1) \delta_2 w^2$$
$$\phi'_1(v_1) = \alpha$$

Then

$$\delta_1 = 0.5 \times 0.1211 \times 0.1 = 0.0061$$

The gradients are then

$$\frac{\partial E}{\partial w^2} = -\delta_2 a^1 = -0.0727$$
$$\frac{\partial E}{\partial w_1^1} = -\delta_1 x_1 = -0.0061$$
$$\frac{\partial E}{\partial w_2^1} = -\delta_1 x_2 = -0.0121$$

3) Updated weights

$$w_1^{1,+} = w_1^1 - \eta \frac{\partial E}{\partial w_1^1} = 1.0006$$
$$w_2^{1,+} = w_2^1 - \eta \frac{\partial E}{\partial w_2^1} = 0.1012$$
$$w^{2,+} = w^2 - \eta \frac{\partial E}{\partial w^2} = 0.1073$$

- 4)** The new output is $y = 0.5161$. In the previous step the output was 0.5150 and the desired output is 1. Therefore, after backpropagation, the weights have been modified to lower the error.

Q2 solution:

a) Induced local field and output (neuron 1 in layer 1)

$$\begin{aligned}v_1^1 &= w_{11}^1 x_1 + w_{21}^1 x_2 = 1.6 \\a_1^1 &= \phi_1(v_1^1) = 0.832\end{aligned}$$

Induced local field and output (neuron 2 in layer 1)

$$\begin{aligned}v_2^1 &= w_{12}^1 x_1 + w_{22}^1 x_2 = 0.9 \\a_2^1 &= \phi_1(v_2^1) = 0.7109\end{aligned}$$

Induced local field and output (Neuron layer 2)

$$\begin{aligned}v^2 &= w_1^2 a_1^1 + w_2^2 a_2^1 = 0.2254 \\a^2 &= \phi_2(v^2) = 0.2254 \\y &= 0.2254\end{aligned}$$

b) Backpropagation. We first calculate the local gradients for each neuron, denoted as δ^2 , δ_2^1 and δ_1^1 .

For the neuron in layer 2, we have

$$\begin{aligned}\delta^2 &= e^{-\phi_2'(v_2)} \\e^{-\phi_2'(v_2)} &= d - y = 0.5 - 0.2254 = 0.2746 \\ \phi_2'(v_2) &= \begin{cases} 1, & v_2 \geq 0 \\ 0, & \text{otherwise} \end{cases}\end{aligned}$$

Then,

$$\delta^2 = 0.2746$$

For the neurons in layer 1, we have

$$\begin{aligned}\delta_1^1 &= \phi_1'(v_1^1) \delta^2 w_1^2 = 0.1398 \times 0.2746 \times 0.1 = 0.1398 \\ \phi_1'(v_1^1) &= \frac{e^{-v_1^1}}{(1 + e^{-v_1^1})^2} \\ \delta_2^1 &= \phi_1'(v_2^1) \delta^2 w_2^2 = 0.2055 \times 0.2746 \times 0.2 = 0.0113\end{aligned}$$

The gradients are then

$$\begin{aligned}\frac{\partial E}{\partial w_{11}^2} &= -\delta^2 a_1^1 = -0.2285 \\ \frac{\partial E}{\partial w_{22}^2} &= -\delta^2 a_2^1 = -0.1952 \\ \frac{\partial E}{\partial w_{11}^1} &= -\delta_1^1 x_1 = -0.0038 \\ \frac{\partial E}{\partial w_{21}^1} &= -\delta_1^1 x_2 = -0.0077 \\ \frac{\partial E}{\partial w_{12}^1} &= -\delta_2^1 x_1 = -0.0113 \\ \frac{\partial E}{\partial w_{22}^1} &= -\delta_2^1 x_2 = -0.0226\end{aligned}$$

c) Updated weights

$$w_1^{2,+} = w_1^2 - \eta \frac{\partial E}{\partial w_1^2} = 0.1228$$

$$w_2^{2,+} = w_2^2 - \eta \frac{\partial E}{\partial w_2^2} = 0.2195$$

$$w_{11}^{1,+} = w_{11}^1 - \eta \frac{\partial E}{\partial w_{11}^1} = 1.0004$$

$$w_{12}^{1,+} = w_{12}^1 - \eta \frac{\partial E}{\partial w_{12}^1} = 0.1011$$

$$w_{21}^{1,+} = w_{21}^1 - \eta \frac{\partial E}{\partial w_{21}^1} = 0.3008$$

$$w_{22}^{1,+} = w_{22}^1 - \eta \frac{\partial E}{\partial w_{22}^1} = 0.4023$$